

NUDGING from AUTOMATIC to REFLECTIVE

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A case study on how behavioral design can be
implemented to improve Barts Health NHS Trust's Waste
Management

With special thanks to:

SKANSKA
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And

Catherine Burrows, Fiona Daly, Joseph Williams, Lazar Armianov,
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ABSTRACT

This thesis explores the theory of nudging and how it can be implemented in Barts Health to improve their current waste management strategies. The purpose of this dissertation is to demonstrate the importance of behavioral design and how it can go hand in hand with hospital waste management in order to enhance the triple bottom line: people, planet and profit.

Incorporating the nudge theory to sustainability strategies is becoming increasingly relevant to companies today, as it has the potential to decrease a multitude of social, economical and environmental concerns we face. The research methodology was a practice based case study comprising of a series of interviews with experts within the field and fieldwork at the Barts Health NHS Trust, in collaboration with Barts Health, SKANSKA and the Carbon Trust. Through this research and inspired by Sille Krukow's case study, I have proposed a nudge to improve waste management and reduce cross contamination in Barts Health.

INTRODUCTION

The definition of nudging is relatively new, having first been coined by Richard Thaler in 2008. Despite its recent definition, the theory has proven itself to be highly effective in encouraging humans to make more sustainable decisions through subtle nudges.

Sille Krukow, a behavioral designer, states:

"Today human behavior is the biggest threat to mankind. We keep overeating, even though we know it's bad for us. We keep using too much energy, even though we know the planet can't keep up with our consumption. We keep driving too fast, even though we know it kills us. But this doesn't mean that we have no intentions to change. We do. But changing actual human behavior calls for good design solutions that take basic human instincts, flaws and habits into consideration. That is design to nudge."

The reason for why there is a gap between how we wish to behave and how we actually act is because humans operate under two systems: the automatic and the reflective. "Biologically we are coded to conserve as much energy as possible, and thus prefer to spend most of our time governed by the automatic system."² This results in passive decision-making, which normally does not benefit us in the long run. With our ever-increasing population, it is imperative to form more sustainable habits in order to address environmental concerns and lack of resources for the over 7 billion people on our planet.

Hospitals are natural candidates for sustainable design as the public buildings place high demands on resources. They have a significant impact on the environment and economy due to their large consumption of energy and water and massive producers of waste. An estimated 40,000 people are hospitalised every day in the UK alone, which results in staggering amounts of carbon dioxide emissions. Carbon dioxide emissions attributable to the NHS in England are greater than the annual emissions from all aircraft departing from Heathrow Airport.³

Although actions have been put in place to reduce carbon emissions from hospitals, at present, there is still little information or guidance available regarding a hospital's visual environment, with specific focus on behavioral design. Being the fifth largest employer in the world⁴, even the smallest changes to the National Health Service can have an enormous impact on the environment due to the massive scale. I chose to focus my research project on Barts Health NHS Trust, the largest Trust in England, as it can act as a microcosm of the NHS. Even within the Trust, which is comprised of 6 hospitals, there are variations between the hospitals and even in the hospitals themselves. This makes the Trust an interesting case study as it provides insights into perspective challenges the NHS might face with incorporating design solutions across various sites.

¹ Design to nudge and change behavior: Sille Krukow at TEDxCopenhagen

² Jespersen, Andreas, and Pelle Hansen. "Nudge Theory: The Mechanics of the Brain." *INudgeyou*. N.p., 03 May 2012. Web. 22 Sept. 2015.

³ Hogg, SKANSKA, Sustainable Waste Management

⁴ Alexander, Ruth. "Which Is the World's Biggest Employer?" *BBC News*. BBC, 20 Mar. 2012. Web. 22 Sept. 2015.

THESIS STATEMENT AND PROBLEM FORMULATIONS

This thesis hopes to explore how implementing a nudge can align with Barts Health NHS Trust's current waste management strategies in order to benefit people, planet and profit.

What barriers does Barts Health NHS currently encounter through their behavioral change strategy and what are some potential solutions to those barriers?

How can behavior design be incorporated to reduce contamination and improve waste management in Barts Health?

METHODS

In order to answer my problem formulations, I have utilized empirical data from both primary and secondary sources. From a three-month work placement at Carbon Trust, I was able to gain insights into statistical data on Barts Health waste management due to the Waste Management Standard awarded by Carbon Trust and their collaboration with SKANSKA. The statistics from the waste management data were useful in gaining an economic perspective as well as scope of how much annual waste the Trust produces and what improvements have already been put in place.

As part of my primary research, I have conducted informal interviews with various experts in the field including Fiona Daly, Environmental Manager of Barts Health, Catherine Burrows, Environmental Waste Manager of SKANSKA, Lazar Armianov, Senior Client Manager of Carbon Trust Certification, Joseph Williams who leads the Carbon Trust behavior change work and Sara Flyvbjerg, Art Director of INDEX: Design to Improve Life® and previous Service Designer of iNudgeyou Development. These interviews were essential in assessing what areas require improvement in Barts Health current waste management strategy, as well as gaining insight into the theory of nudging and behavioral design.

In order to understand the current waste management policies and practices at Barts Health I was fortunate enough to gain access to Whipps Cross Hospital for fieldwork, in which I shadowed a waste management training session as well as an auditing process. During my fieldwork, I paid particular attention to the placing of bins, the labeling and user-experience. It was vital to focus on the user, that being the staff, in order to understand their perception of medical waste management training and the difficulties they face during the current waste segregation process. In order to determine if there was a staff-driven interest in sustainability and to ascertain what barriers they encounter during their day-to-day practice with regard to waste disposal, I also applied a survey (Appendix 1). Through mapping my user research I was able to identify the unique challenges faced by Barts Health.

In addition, I have also utilized external literature provided by SKANSKA and Carbon Trust as secondary sources to support my primary findings. From analysis of both primary and secondary research I was able to design and prototype a potential solution by applying the nudge theory.

I have analyzed the solution scenario by applying a PESTEL model, in order to discover what Political, Environmental, Social, Technological, Economical and Legal implications my solution scenario has from a macro-environment perspective.

Finally, I have applied a stakeholder analysis in order to discover who the key stakeholders are, their stakes in the company and how they will be impacted by the strategy. I selected a stakeholder analysis as it evaluates what the value propositions are and therefore draws attention to the benefits of the strategy. This has assisted in the conclusion of my thesis by outlining how the solution might help solve some of the barriers Barts Health faces through their current waste management strategy.

DELIMITATION

Although the improvement of waste management is important throughout the entire NHS, I have selected to focus my research dissertation on Barts Health NHS Trust due to the geographic location, their size, as they are the largest Trust in England, as well as the massive strides they have taken in improving sustainability through behavior change in recent years.

This makes it a particularly interesting case study to analyze as it provides guidelines for how other Trusts could improve their waste management, and to see how implementing a nudge could spur them towards achieving their goal of being the most sustainable trust by 2020.

A limitation I encountered during my research is that of the 6 hospitals of Barts Health, I was only able to obtain access for fieldwork at Whipps Cross Hospital. I have therefore selected this hospital to be the 'pilot site' for the prototype, which is fitting, as it is one of the least advanced in terms of waste management out of the 6 hospitals. Under the timeframe and due to the varying nature of the hospitals, it was more valuable for me to select one to focus on in more detail. The bureaucratic nature of hospitals and limited access is why I have only discussed a broad overview of the suggested implementation for my solution scenario; the finer details of the strategy are not essential to answering my problem formulation. Ideally, I would have tested my prototypes in their intended settings to assess their effectiveness, but administrative barriers meant it was beyond the scope of this thesis.

I have chosen not to include any cultural probes or forecasting/backcasting, as with nudging it is imperative to have a more scientific approach with statistics and measurements. This is another reason why I have focused on waste management, which can be measured, as opposed to something more subjective like mood and emotions, which are not easily measured against an original baseline.

RESEARCH

BARTS HEALTH NHS TRUST

Formed in 2012, through the merger of the Barts and The London NHS Trust and three other London-based NHS Trusts, Barts Health is the largest NHS Trust in England. Barts Health NHS Trust comprises of 6 hospitals - Royal London, The Chest, St Bartholomews, Newham, Whipps Cross and Mile End.

With an annual turnover of over 1.25 billion pounds, the Trust employs 15,000 staff and delivers healthcare to over 2 million patients a year (footnote). Barts Health has ambitious goals of becoming the most sustainable trust and reducing their impact by 15% per patient by 2020.⁵

Although the Trust is a massive producer of waste, Barts Health has already made massive strides in improving their waste management processes. The Trust, which disposes of 6,000 tones of waste a year, has eliminated all of its waste from landfill since October 2013, increased recycling by 384% and saved the NHS over 700,000 pounds.⁶ These efforts have not gone unnoticed, as Barts Health was the first NHS Trust in England to receive the Carbon Trust Waste Management Standard. The standard aims to focus organizations to examine their resource use and environment impact, evaluating and measuring their use of the waste hierarchy.

SKANSKA PARTNERSHIP

These impressive results can be attributed to the drive towards cultural change within the staff and educating the importance of correctly segregating the waste streams. This has been achieved through the partnership with waste contractors, SKANSKA and the implementation of a new behavioral change program. Over the past two years, more than 5,000 staff members have been trained on waste segregation and over 180,000 individual bin audits have been carried out, resulting in improving compliance by almost 400%.⁷

There is clearly good communication between the Trust and SKANSKA. Fiona Daly, Environmental Manager at Barts Health NHS Trust, said:

"It was important for us, when tendering the waste management service, that we chose a partner who fully aligned with our values, drove innovation and bought into our vision for a sustainable healthcare future. Our partnership with Skanska has already delivered excellence in data collection and reporting, quality assurance, compliance improvements and innovative solutions."

⁵ *Barts Health Environmental Sustainability Strategy Our Vision for a Sustainable Future*. Publication. N.p.: Barts Health, n.d. Print.

⁶ "Sustainability." *Barts Health*. N.p., n.d. Web. 22 Sept. 2015.

Sust-N Ltd Waste Report Barts Health NHS Trust. Rep. N.p.: SKANSKA, 2013. Print.

⁷ *Barts Health Environmental Sustainability Strategy Our Vision for a Sustainable Future*. Publication. N.p.: Barts Health, n.d. Print.

COMMODITIZING WASTE

The Trust has clear financial incentive to keep waste to a minimum as it pays for the actual amount of waste processed. Previously the Trust contracted on the basis of thresholds, so there was no extra cost provided the waste fell within certain parameters. In addition, the Trust is at present in 93 million pounds deficit meaning that there is an increased motivation to save money from all corners.⁸

The Trust was able to achieve a zero waste landfill through redirecting the domestic waste to a Refuse Derive Fuels site for re-use into fuel pellets and all paper and cardboard waste is segregated and recycled, among other redirections of waste streams.

SKANSKA and Barts Health continue to seek new, innovative ways of commoditizing waste. The segregated waste which can be sold for recycling, such as the cardboard is now being "commoditized" as they are extracted and used to raise money. Mr Hogg, says "Instead of paying someone to take it away, someone is paying us."⁹

Other innovative steps have been implemented through for example installing the solar bins and the first Re-vending machine into the NHS.



⁸ Boseley, Sarah. "Barts Health Chief Executive and Chief Nurse Resign." *The Guardian*. The Guardian, 19 Feb. 2015. Web. 22 Sept. 2015

⁹ An HSJ Supplement/4 October. "Throwing the Old Ways in the Bin." *A Closer Look at Partnership* (2013): 4-5. HSJ, 4 Oct. 2013. Web. 22 Sept. 2015.



IMAGE: Big Belly Solar Bins



IMAGE: Re-vending machine

The team at Barts Health NHS Trust is working to create an exemplar case study for the sustainable management of healthcare waste and hope to support other Trusts to deliver similar results.

Jane Sansome, Business Director at SKANSKA UK commented:
*"It's through a close working relationship and the sharing of ideas that we have been able to make such significant strides in the way that waste is managed, which is cheaper, greener and contributes to creating the best possible environment for patients."*¹⁰

BACKGROUND OF BEHAVIORAL CHANGE PROGRAM

The behavioral change contract put in place by SKANSKA involves interfacing with 15,000 staff, 2 million patients and visitors cross the 6 hospitals. Changing behaviors is a key part of the Trust's sustainability agenda moving forward as ensuring staff integrates effectively with their built environment improves patient experience and achieves efficiencies. The 'changing lives campaign' and awareness programs have had enormous benefits to the triple bottom line: people, planet and profit.

I met with Catherine Burrows, Environmental Waste Manager at SKANSKA, in order to find out more about the behavioral change program and its background.

"The critical success factor was being able to influence the clinical staff. We knew that sending emails or other written communication would have little effect. Instead, we established a small team that would visit the clinical departments, talk to staff members and try to change their behavior. We were amazed at how readily people embraced the concept. We started to identify "champions" and they were acknowledged by the trust's chair when he presented certificates. This had a knock-on effect as more staff members became committed to appropriate waste disposal." says Burrows.¹¹

The origins of the behavioral change team came when an autoclave (sterilizer) was put in place to process clinical waste. Liam Hogg, SKANSKA's then appointed Waste Manager, who put the program in place said: "Previously we emptied the waste into the incinerator and all you got out was black ash. But with the autoclave we could see what came out the other end, and there were all sorts of things, like chocolate boxes and daffodils. So yes, the NHS was paying to sterilise bunches of flowers."¹²

Due to the stressful environment of the hospital setting, poor placement of bins and lack of training, too often the default position for busy hospital workers would be to throw everything into the clinical waste bin. The program's aim is to facilitate segregation of waste for staff through a mélange of behavior change techniques and adaptations to the physical working environment.

According to Hogg, solving just one major issue – incorrect disposal of clinical waste – could save the NHS £40m per year. "Around 60

¹⁰ An HSJ Supplement/4 October. "Throwing the Old Ways in the Bin." *A Closer Look at Partnership* (2013): 4-5. HSJ, 4 Oct. 2013. Web. 22 Sept. 2015.

¹¹ "Catherine Burrows, Environmental Waste Manager SKANSKA." Personal interview. 20 Aug. 2015.

¹² An HSJ Supplement/4 October. "Throwing the Old Ways in the Bin." *A Closer Look at Partnership* (2013): 4-5. HSJ, 4 Oct. 2013. Web. 22 Sept. 2015.

per cent of 'clinical waste' is actually domestic waste," he says. "And given that disposing of clinical waste costs two, three or even more times than it costs to dispose of domestic waste then we're talking big potential savings."¹³

Mr Payne, who created NHS Sustainability Day and who is director of estates and facilities at Barts Health Trust in London, says that good waste management is not only right for the environment, it also gives a welcome boost to the financial bottom line.

"It's not a rocket science issue, it's not sexy, but waste management is a big problem for every trust and every health service site," he says. "It's important to realize that getting it right isn't a 'nice-to do', it's a must."¹⁴

Improper waste management is an issue for plenty of Trusts across the NHS. According to the NHS Sustainable Development Unit (SDU), "inefficiently managing waste costs the NHS money that could otherwise be spent on direct patient care". Once the results from training and auditing showed how it saves money for frontline patient care and protection of the environment, the staff became far more passionate about the reasons why the behavioral team was put in place. Karen Hogg, Operations Manager for SKANSKA's subcontractor SUT-N said, "I've really noticed the difference since we started out," she says, explaining that it is now much rarer to find clinical waste in the domestic bin. Letting people know that domestic waste is actually recycled (rather than going to landfill) has been a real means of encouragement."¹⁵

THEORY

The behavioral team currently uses the 'hug' method. The Trust presents certificates to the most sustainable departments and winners also receive a prize pack, clearly creating incentive to segregate waste appropriately. As staff receives rewards and the decision to separate waste is more of a conscious decision, it falls into 'Hug' rather than Smack, Shove or Nudge¹⁶.

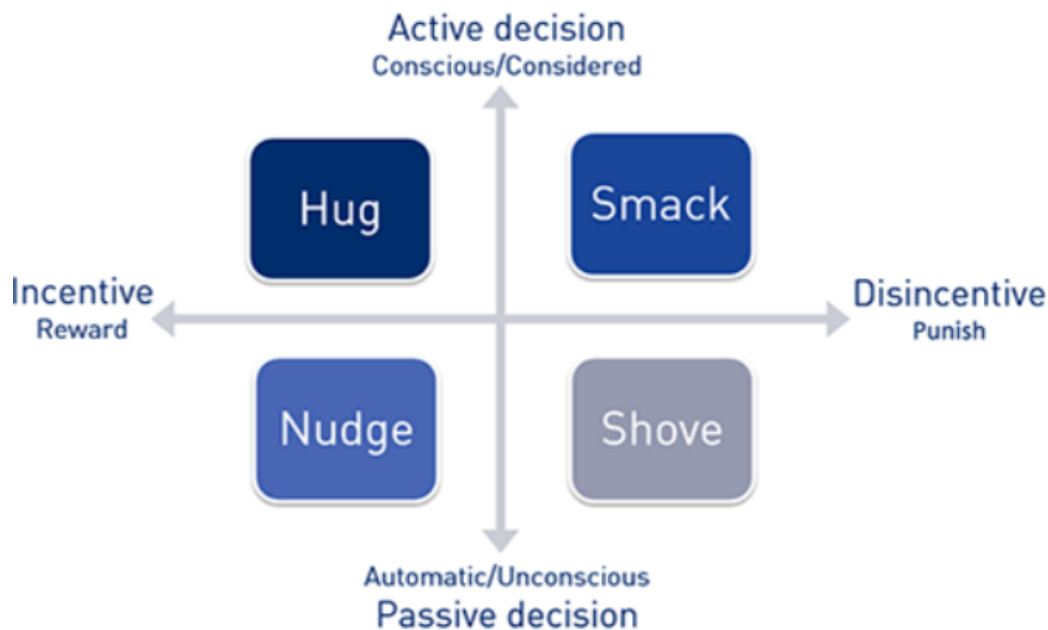
¹³An HSJ Supplement/4 October. "Throwing the Old Ways in the Bin." *A Closer Look at Partnership* (2013): 4-5. HSJ, 4 Oct. 2013. Web. 22 Sept. 2015.

¹⁴An HSJ Supplement/4 October. "Throwing the Old Ways in the Bin." *A Closer Look at Partnership* (2013): 4-5. HSJ, 4 Oct. 2013. Web. 22 Sept. 2015.

¹⁵"Sustainability." *Barts Health*. N.p., n.d. Web. 22 Sept. 2015.

Sust-N Ltd Waste Report Barts Health NHS Trust. Rep. N.p.: SKANSKA, 2013. Print.

¹⁶Image: <http://www.carbontrust.com/resources/tools/empower>



HUG - Covers a range of incentives for positive decisions.

NUDGE - Based on the idea of changing behavior without using compulsion.

SHOVE - More deliberate than a nudge. F.ex. Restrictions of what types of bins you are allowed to use, like coding them.

SMACK - Banning, punishment.¹⁷ F.ex. Fine for incorrect segregation.

NUDGING

The "Nudge" theory was coined by economist Richard Thaler in 2008 with the release of his book "Nudge: Improving Decisions About Health, Wealth and Happiness", written with Cass Sustein. Their definition is: "A *nudge*, as we will use the term, is any aspect of the choice architecture that alters people's behavior in a predictable way without forbidding any options or significantly changing their economic incentives."¹⁸

Far before Richard Thaler defined the term, however, examples of nudging have altered behaviors in history. An early example was put in place in 1919 when an American highway in Michigan had a white line put down the middle of the road in order to avoid car collisions.¹⁹ Something that has been prominent throughout all these examples is their simplicity.

The Nudge theory follows a hybrid approach to influencing human behavior as it is rooted in a combination of Behavioral Economics, Psychology and Political Theory. A classic nudge follows three traditions in psychology, as outlined by Dr David Halpern, CFO of the

¹⁷ <http://www.bbc.co.uk/news/health-21593335>

¹⁸ Thaler, Richard H., and Cass R. Sunstein. *Nudge: Improving Decisions about Health, Wealth, and Happiness*. New Haven, CT: Yale UP, 2008. Print.

¹⁹ Example given by Dr Halpern at LSE lecture on Nudging held on 15 September 2015

Behavioral Insights Team, at the LSE lecture on "Inside the Nudge Unit: how small changes can make a big difference"²⁰:

1. Experiential and Perceptual e.g. looking for change, contrast, edges
2. Social Psychology e.g. how we behave and interact with each other
3. Cognitive Psychology e.g. how people think and make decisions

Nudging aims to create predictable behavioral outcomes through the premises of how the human brain perceives the world and makes decisions.²¹ The foundation of the nudge theory is based on a phenomenon known as Dual Process Theory (DPT). This is the separation between the automatic and reflective systems in our brain. As stated in the introduction, humans are primarily set in the automatic system as it conserves the most energy. This system controls most of our behavior, however, those behaviors are often unconscious, fast and associative. In essence, this system is the 'auto-pilot' mode.

The second system, reflective, follows the opposite approach, as it is conscious thought, slow, analytical and consumes a great deal of energy. The reflective system requires more energy as it works by creating hypotheses based on prior experiences and knowledge before acting. This system can be activated by will, but it takes a lot more consideration and can only process a fraction of alternatives compared to that of the automatic system. This is why the reflective system tends to only be activated when presented with an odd, rare or potentially dangerous situation. It has to be prompted and 'awoken', so to speak.

These two systems work side-by-side, aiding humans to conserve energy and make decisions. There are three main factors, however, that cause rifts in the partnership:

- The reflective system remains unaware of the problem
- The automatic processes provide us with the wrong response
- The reflective system provides us with the wrong response

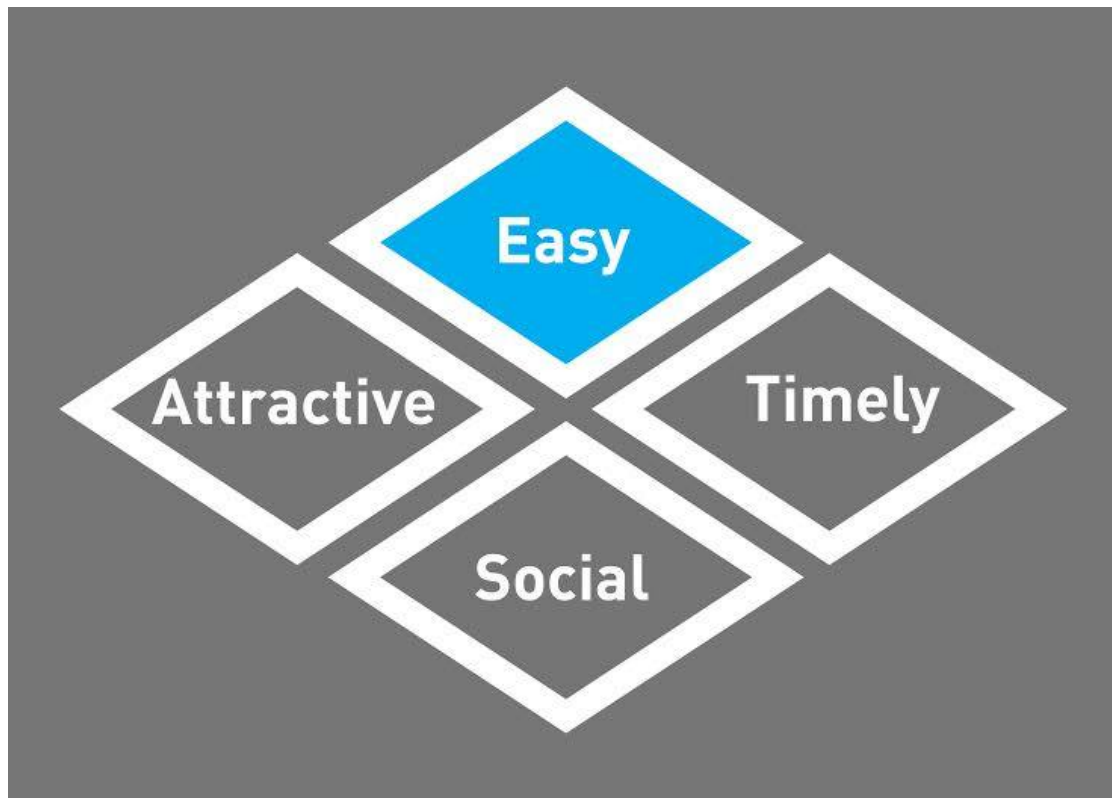
In terms of the hospital environment, the staff is often forced into the reflective system as they encounter dangerous situations, puzzling events and stress that requires them to be alert in their day-to-day. Unsurprisingly, this has a knock-on effect on less life-threatening situations, like hospital waste management, where the automatic system is switched back into place to conserve precious energy sources.

The nudge works by countering the wrong automatic responses by gently guiding users in their surroundings to make more sustainable decisions. Often nudges are used to make our automatic system choose differently, without even being aware of it, which is why

²⁰ Halpern, David, Dr. Inside the Nudge Unit: How Small Changes Can Make a Big Difference. LSE, London. 15 Sept. 2015. Lecture.

some critics perceive nudging as manipulative. Other times, nudging is aimed at activating our reflective system at the right time.²²

The guidelines for designing a successful nudge is to follow EAST:



EASY

The first, and arguably most important, aspect of a nudge is to make what you want to achieve as easy for the user as possible. The more obstacles the user has to overcome to make a decision, the less likely they are to complete the act. This holds true in matters both large and small. An example, which was highlighted at the London School of Economics lecture on nudging held by Dr Halpern, included the dramatic drop in suicide rates in the UK after the British government phased out coal gas for less lethal natural gas ovens. Prior to this swap, oven-assisted suicides accounted for half of the suicides in the UK. Once the natural gas ovens were put into place, the suicide rate in the UK dropped by a third.²³

²² Jespersen, Andreas, and Pelle Hansen. "Nudge Theory: The Mechanics of the Brain." *INudgeyou*. N.p., 03 May 2012. Web. 22 Sept. 2015.

²³ Hersh, Julie. "Can an Obstacle Prevent Suicide?" *Psychology Today*. N.p., 02 Dec. 2010. Web. 22 Sept. 2015.

ATTRACTIVE

Another important element of nudging is to make the decision attractive for the user. This could mean visually attractive, fun to do, or to peak their attention in some way. An example of this includes the Piano Stairs in Stockholm that were put in place by Volkswagen to encourage people to take the stairs instead of the escalator. The steps were installed with motion-sensor piano keys, which meant that they played musical tunes when they were climbed. The stairs appealed to their users by making them eye-catching and interactive. After only a one-day trial, results showed that 66% more people took the stairs rather than the escalator.²⁴



SOCIAL

We as humans operate 90% based on instincts, and much of that follows the pack mentality. Behavioral scientists have found that one of the most effective methods for behavioral change is to harness people's tendency to follow social norms. This can be done through a 'herd-mentality nudge' and consensus messaging. An example of this includes a study conducted to compare the effectiveness of messaging to get hotel guests to reuse their towels. The message which was framed in terms of social norms: "the majority of the guests in this room reuse their towels" proved to be more effective than the one that highlighted the environmental benefits of reuse.²⁵

TIMELY

The last component of a successful nudge is to be timely. A well-timed nudge, i.e. at times when it's most convenient for the user to be prompted with an option, has been proven to be highly effective. An example of this includes offering vouchers for unsubsidized fertilizers shortly after harvest when farmers have more cash at hand. This "nudge for development" was proposed to be implemented by the Malawian government in order to improve maize farming in a sustainable and cost-efficient manner.²⁶

²⁴Kirkup, James. "Musical Stairs Could Nudge Britain to Better Health." *The Telegraph*. Telegraph Media Group, 31 Dec. 2010. Web. 22 Sept. 2015.

²⁵Sleek, Scott. "Small Nudge, Big Impact." *Association for Psychological Science* RSS. N.p., n.d. Web. 22 Sept. 2015.

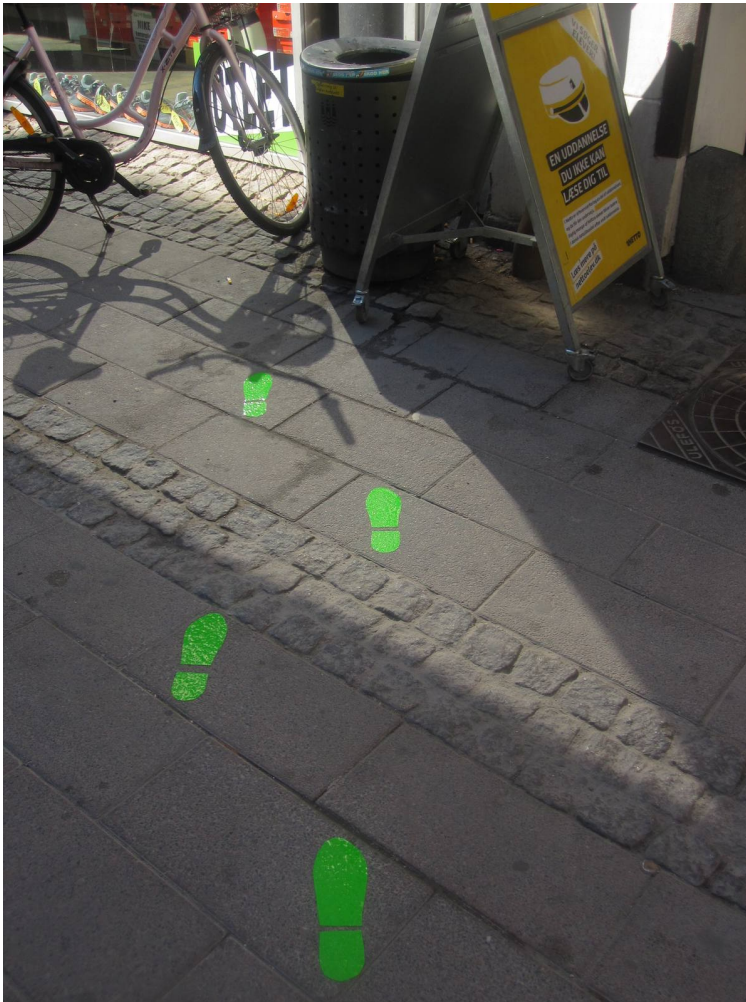
²⁶"Timely Nudges Could Address Malawi's Increasing Agricultural Woes." *Innovations for Poverty Action*, 17 Oct. 2011. Web. 22 Sept. 2015.

CASE STUDIES

GREEN FOOTPRINTS COPENHAGEN

Sara Flyvebjerg, Art Director of INDEX: Design to Improve Life® and previous Service Designer at iNudge, highlights that an important element of a successful nudge is that it can be measured against an original baseline²⁷. Flyvebjerg illustrates this by mentioning the litter-reducing nudge that was implemented in Copenhagen. Despite the fact that a survey showed that Danes claim to be concerned about littering, a study showed that every third Dane occasionally leaves trash in streets or parks. This is not quite that surprising as there is often a gap between attitude and behavior.

Developed in 2011 by Roskilde University and iNudgeyou, green footprints were put in place throughout the streets of Copenhagen in order to lead pedestrians to bins.



In order to test the whether this nudge was efficient in reducing waste or not, the students from Roskilde University conducted an experiment using caramel wrappers. In the first test they handed

²⁷ "Sara Flyvebjerg, iNudgeyou." Telephone interview. 22 July 2015.

caramels to pedestrians and then counted how many wrappers were placed in the bins, the street and bicycle baskets. In the second test, they handed out the caramels again but this time had put the green footprints in place. This resulted in a 46% reduction of wrappers in the streets.²⁸

MAKE IT EASY

Part of the reason the green footprint nudge is so effective is because it follows the nudge-doctrine of making things easy. By guiding the way in busy streets, the nudge is efficient in closing the gap between what people initially want and how they should behave. *"Littering is often caused by bad habits. Even if there were an economic penalty for littering, usually some sort of fine, it would need to be enforced daily and everywhere to really impact people's habits and automatic behavior. This isn't exactly the cheapest solution. The footprints, on the other hand, work as a visible reminder for pedestrians who aren't fully aware of their actions when they litter and so end up being a far more effective tool than empty threats. Lastly, it's a sign of respect, and a recognition of the fact that the citizens of Copenhagen actually do care about the state of the city streets, they just need a little nudge to effectively carry those concerns out in their busy every day lives."*²⁹

This example applies to waste management in Barts Health, as there is a gap between what people want and how they act, which I found from my survey and from shadowing the auditing at Whipps Cross Hospital, which I will explain in more detail further on.

SILLE KRUKOW

Sille Krukow, a member of iNudgeyou and The Danish Nudging Network, is the Chief Behavioral Designer at KRUKOW, a behavioral design consultancy developed in 2008.³⁰ Krukow has been involved in numerous waste management projects in Denmark, including the green footprints mentioned above as well as a nudge put in place for the municipality of Odsherred and Hold Danmark Rent.

The project is an excellent case study on how behavioral design can be implemented to reduce contamination and improve waste management. In her Tedtalk³¹ she outlines a study done on a Danish beach, where she conducted a one-week experiment using color systems, strategic placements and visual cues to nudge a better sorting of waste. Prior to putting the nudge in place, Krukow conducted fieldwork to observe how people went about throwing away trash at the site. More often than not, people simply placed all glass, metal, and plastic in one bin or left trash strewn across the beach.

"If we don't have a visible trashcan then we are more likely to leave trash in the sand. Not because we are lazy or don't care, but because

²⁸ "Nudge Cases - Sille Krukow." *Sille Krukow Nudge Cases Comments*. N.p., n.d. Web. 22 Sept. 2015.

²⁹ <http://inudgeyou.com/green-nudge-nudging-litter-into-the-bin/>

³⁰ <http://sillekrukow.com/>

Krukow, Sille. "Design to Nudge and Change Behaviour: Sille Krukow at TEDxCopenhagen." TEDxCopenhagen. Copenhagen. 4 Oct. 2013. Lecture.

our attention is directed elsewhere. I rearranged the trashcans so that they were clearly visible, secondly, I gave the trashcans different colors, so users could intuitively understand that there is some kind of system going on and certain behavior expected of us." Krukow stated.³²

This nudge had a tremendous positive effect and resulted in 90% clean waste fractions that were ready to recycle.



Krukow also notes in her TedTalk that demonstrating what you want people to do has a much better effect than telling them what not to do. She explains that humans have many flaws including limited attention and that our behavior is reliant on our surroundings. Our instincts are so powerful that the majority of the time we are set in

³² Krukow, Sille. "Design to Nudge and Change Behaviour: Sille Krukow at TEDxCopenhagen." TEDxCopenhagen. Copenhagen. 4 Oct. 2013. Lecture.

the automatic frame of mind, rather than the reflective. This is why creating solutions that guide users is the most successful strategy, as opposed to relying on simply telling/training people once.

TLC PROGRAM

Aside from the behavioral program put in place by Barts Health, the Trust has also implemented the nudge philosophy through their TLC program. The TLC (Turn Off - Lights Out - Close Doors) project was trialed in two hospitals: St Bartholomew's Hospital and The Royal London Hospital. The pilot scheme involves saving energy and money by persuading staff to turn off lights, switch off machines and close doors - much like the name suggests.

The inspiration, practical support, and funding for the project came from a partnership between Barts Health NHS Trust, General Electric, SKANSKA and Global Action Plan. The partnership worked with the Trust for more than a year in order to plan and design the behavior change program or 'nudge'.

"Sometimes getting people to do simple things can be a complex task, especially in the NHS where focus is rightly on patient care and not light bulbs. But Sustainability programs can contribute to an organization's core objectives." said Global Action Plan's CEO, Trewin Restorick.³³

Barts Health executed the project by appointing sustainability champions among matrons and other ward staff who then encouraged colleagues to change their behavior and become more environmentally friendly. Rewards such as pens, biscuits and fair-trade snacks were also used to get staff to join in the challenge.

Results and estimated savings come from data collected by the partnership based on behavioral audits, patient experience surveys and energy data, and were independently evaluated by the Clinical Research Centre at Queen Mary University London. As well as the financial savings, patients also benefited with many reporting better quality sleep, a third fewer incidences of sleep disruption than in non-pilot wards by doors being shut and privacy increased.

Barts Health NHS Trust Environmental Manager, Fiona Daly, said: *"Operation TLC has delivered improved patient care at the core of our business whilst empowering our entire workforce to take action on energy usage and climate change. It has been a truly inspiring project and one I hope other NHS Trusts will be inspired to follow. If similar "Nudge" techniques were used across all six of our hospitals we could save a further £400,000. We also estimate if all health trusts in the UK introduced these techniques, the NHS could save as much as £35million in a year."*³⁴

³³ Barts Health NHS Trust. *Ground Breaking Behavioural Techniques Forecast to save the NHS's Largest Trust £400,000 over the next Year*. N.p., 24 June 2013. Web. 22 Sept. 2015.

³⁴ Barts Health NHS Trust. *Ground Breaking Behavioural Techniques Forecast to save the NHS's Largest Trust £400,000 over the next Year*. N.p., 24 June 2013. Web. 22 Sept. 2015.

The use of incentives classifies the project as a 'hug' rather than a 'nudge' in my opinion. It is also my opinion that a leading factor for the success of the program was not so much impacted by the physical rewards, but rather demonstrates that raising awareness of patient-benefits is a key incentive.

This example also illustrates that despite a behavioral change project being effective in one Trust, the experiments tend to terminate after trials. This makes me question, why if such cost-efficient behavioral change programs has been proven successful why they are not scaled to other Trusts, let alone even continued.

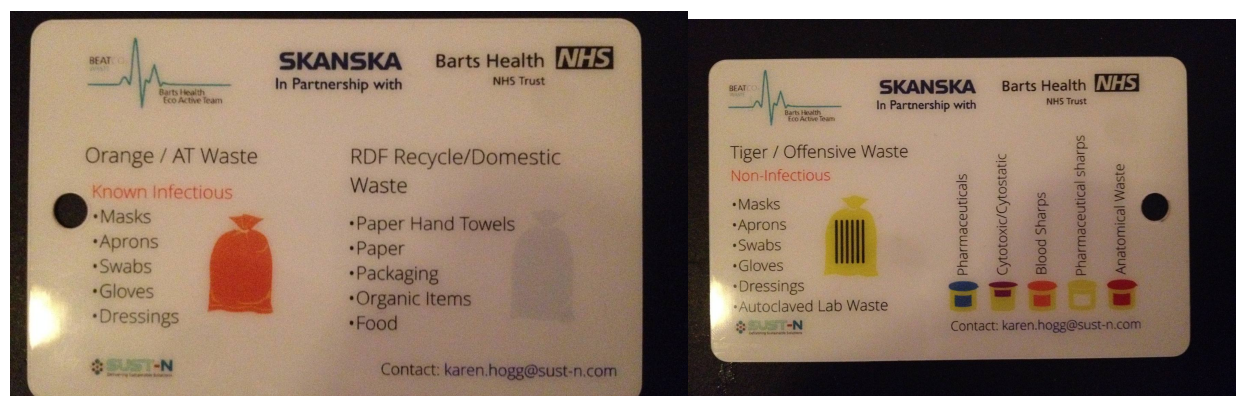
FIELDWORK

Whipps Cross Hospital



I was fortunate enough to arrange a visit at Whipps Cross Hospital to observe first hand how SKANSKA's behavioral team operates. As part of my fieldwork, I shadowed Nina (a waste management auditor) during her waste management training and auditing. There were approximately 10 nurses present at the training, all whom had never had training before despite working at the hospital for a significant amount of time. Nina began the training by explaining the reasoning for the different bins and why hospital waste has to be segregated. This is due to law. The driving factor, however, is about cost as the more money saved through improved waste management comes directly back to the Trust.

After this Nina distributed cards (which you can see below) which outlines the different waste in the hospital:



TYPES OF WASTE



Offensive Waste
18 01 04
Non infectious



Alternative Treatment
18 01 03
Infectious Clinical Waste



High Temperature Waste
18 01 08, 18 01 09, 18 01 03



Medicinal Sharps



Medicinal Products



Anatomical Waste



Blood Sharps



Cytotoxic and Cytostatic Medicines

There are principally two ways of dealing with clinical waste, one is to treat with high temperature steam (“autoclaving”) and the other is to dispose by high temperature incineration. Clinical waste has to be separated into different containers for disposal. The type of container conveys how hazardous or infectious the waste might be. A color-coding system has been developed to indicate to both the transporters and disposal contractors just what is in each container.

- **AT Waste: Known Infectious**
Waste which may be “treated”. Indicative treatment/disposal required is to be “rendered safe” in a suitably permitted or licensed facility, usually alternative treatment plants (ATPs). However this waste may also be disposed of by incineration. Ex. masks, aprons, swabs, gloves, dressings
- **Tiger/Offensive Waste: Non-Infectious**
Waste which requires disposal by incineration. Indicative treatment/disposal required is incineration in a suitably permitted or licensed facility. Ex. masks, aprons, swabs, gloves, dressings, autoclaved lab waste
- **RDF Recycle/Domestic Waste.**
Minimum treatment/disposal required is landfill, municipal incineration/energy from waste or other municipal waste treatment process at a suitably permitted or licensed facility. Recyclable components should be removed through segregation. Clear/opaque receptacles may also be used for domestic waste. Ex. paper hand towels, paper, packaging, organic items, food
- **Pharmaceuticals**
Indicative treatment/disposal required is incineration in a suitably permitted facility.
- **Cytotoxic/Cytostatic**
Indicative treatment/disposal required is incineration in a suitably permitted or licensed facility.
- **Blood Sharps**
Indicative treatment/disposal required is incineration in a suitably permitted facility.
- **Pharmaceutical Sharps**
Indicative treatment/disposal required is incineration in a suitably permitted facility.
- **Anatomical Waste**
Indicative treatment/disposal required is incineration in a suitably permitted facility.

The correct segregation of healthcare waste onsite is vital to ensure that waste is stored, transported and ultimately disposed of in the correct manner to comply with clinical waste regulations.

If infectious waste is disposed of incorrectly a department could incur up to a 10,000-pound fine. Nina outlined the consequences of cross contamination - due to the cable ties on the trash, there are codes that can trace back to the original department where it was disposed of, meaning that someone is liable.

This is due to the risk of infection of specific waste; Nina explained that just in the first 10 days of September there were 3 different needle stick injuries at The Royal London Hospital. The risk of injuries, which is clearly still high, in a hospital environment is a crucial factor of why waste management must be improved.

Other waste is of course extremely sensitive, such as the anatomical waste for example. It is imperative that once a body part has been disposed of in the red bin, it is removed off-site within 72 hours by law. If the lid is dated at the wrong time, they could suffer consequences by law. To the shock of the nurses, they discovered they had been sorting the sharps and medicines wrongly and had not even known some of the bins i.e. that the pharmaceuticals bin existed. Nina said she would order more bins to be placed as well as more tiger waste bin bags as that department had run out.

I discovered something particularly interesting during my primary research of the training. One member of staff inquired why there was not clearer labeling on the bins, to which Nina answered that the Trust is not in favor of adding icons due to the risk of change in procedure. For example, previously food was not possible to recycle and now it is. However, she also explained that individual departments could add whatever labels they wish as long as they abided by protocol and were laminated.

This would be useful, as some staff show disregard to standard waste management procedures when unsupervised and this runs the risk of the whole department being fined. Nina noted, "Would you like this coming out of your wages?" The financial consequences of incorrect waste segregation are enormous as orange/known infectious waste (380 pounds a ton) costs more than double to process compared to Tiger waste (160 pounds a ton).

AUDITING

The auditing initiative involves deploying a specially trained set of workers once a month to audit the contents of clinical and other bins to ensure that waste is being segregated properly. The results of each individual bin audit are recorded on iPads and then stored on a database. It is here that managers get an overview of how every department is meeting waste segregation guidelines. Over the past 4 years, the team has conducted over 250,000 bin audits. The team records if there is any miss segregation and highlight where more training is needed to improve compliance.

Alongside side this; the new program has considered the placement of bins to make it easier for staff to dispose of waste. "If someone has to walk to bins in two different places to segregate waste they are less likely to make the effort, so we give a choice at the point of disposal," explains Rachael Baldwin, sustainability manager with SKANSKA. "People are busy and have other priorities at the time. So we make it easier to do the right thing by putting the bins together, and everyone wins."³⁵

It was my aim to shadow Nina in order to see if the previous statements held true. Going around the different wards, I noticed some recurring problems. One of the most notable issues was the positioning of the clinical waste bin and the domestic waste bin. In the image below you can see that the clinical waste bin is closer to the sink and is actually blocking access to the tissues. This demonstrates that while both bins are present and fulfill that

³⁵ "Skanska Waste-reduction Measures to Cut Costs for Barts Health NH - Skanska." *Skanska*. N.p., 19 Dec. 2013. Web. 22 Sept. 2015.

standard, placement of them is extremely important and should not be neglected.



Nina explained that it is not part of her role as auditor to rearrange the bins more strategically, but merely to check if the bins are in the individual rooms and if the contents are sorted inside of them.

In the following image, you can see a more strategic positioning of the bins as the domestic is closer to the sink. Throughout all the departments, however, none of the bins had any labeling aside from the black stripe, which made it difficult to distinguish the different kinds of clinical waste.



In the below image, you can see that a medical glove was disposed incorrectly into the domestic waste. From our audits, this was the case in the majority of the domestic waste bins - a clear indication that efforts still need to be put in place to improve waste segregation.



In the image above you can also see that the domestic waste bin is overfilled, which could prompt staff or visitors to dispose of their trash in a clinical waste bin instead.

Although I was not able to gain access to any of the other hospitals that are part of Barts Health, I was able to receive images of bins at the Royal London Hospital courtesy of Dr. Sara Dahlén:



As you can see in the image above, some of the bins in the Royal London Hospital have labels on them indicating whether they were Clinical Waste or General Waste, rather than simply the black line. However, from the images I received, positioning of the bins seemed to have consistent problems. As you can see with the clinical waste bin being difficult to access:



In addition, where there are labels missing, it is difficult to distinguish between Tiger bins (Yellow striped) and Known infectious waste (orange):





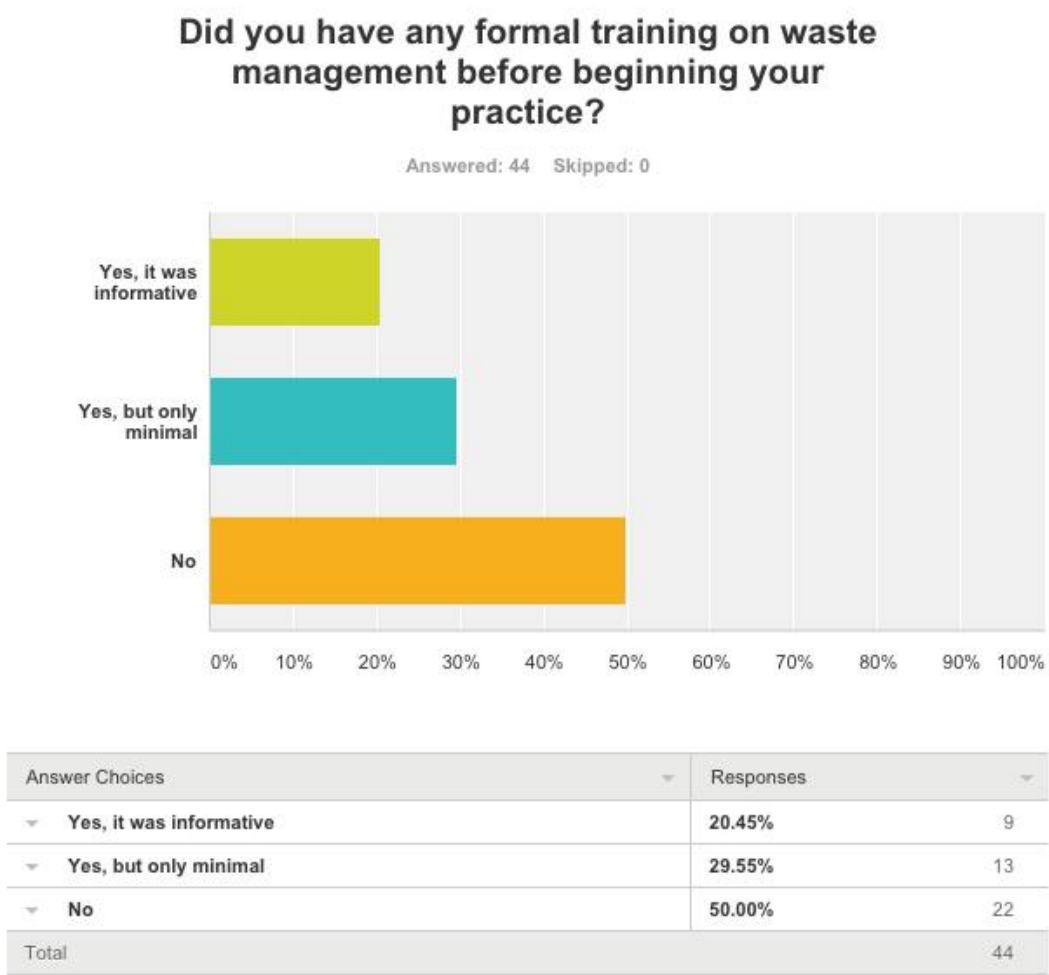
From my observations, the main barriers Whipps Cross and The Royal London Hospitals face in waste management are:

- Inappropriate location of waste bins
- Varying sizes of bins
- Waste bins being overfilled
- Lack of strategic placement of bins
- Lack of knowledge about the constituents of medical waste
- Lack of labelling
- Lack of bins
- Similar features of bins despite various purposes

SURVEY

Upon completion of the audit, I set off to find staff members to complete my questionnaire (See Appendix). The aim of my questionnaire was to gain information on user-research and insights into what barriers staff face in hospital waste management. I broadened the range of my survey to not only include staff in Whipps Cross but also staff from across Barts Health's sites and elsewhere. From the 44 respondents, the most prominent issues that were highlighted included lack of formal training before beginning their practice, discrepancies with either too many bins or too few, the need for clearer labeling and more strategic placement of the bins.

When asked about training prior to beginning their practice, an astounding 79.55% responded that they either had no formal training or the training was minimal.



When questioned "What are the greatest difficulties for you in terms of disposing of waste? How do you feel waste management could be improved/facilitated for you?" I received contradictory responses:

- "More bins and different colours"
- "More waste bins"
- "More bins and labeling"
- "Finding a bin, increase number and make them easy to find."
- "There are too many bins, reduce them."
- "Finding the right bin. More visible marking. More education/training"
- "Icons of what goes into the bin"

:Finding domestic waste bins that are clearly labelled for recycling"

"Ensuring patients put waste in correct bins, they just seem to put waste in whatever bin is nearest. Maybe even clearer labeling and colours."

"Availability and increased variety of bins in all areas"

"Clearer labeling"

This was important for establishing user-needs as the staff is the primary users of the bins.

On a positive note, when asked how important recycling in hospitals is to them, from 'Not important', 'Do not consider it', 'It's great if it happens but otherwise not a concern', 'Reasonably important' and 'Incredibly important' A very notable 47.73% answered 'Incredibly important', 38.64% answered 'Reasonably important', only 13.64% answered 'It's great if it happens but otherwise not a concern' and 0% answered that recycling was not a concern.

	Not important	Do not consider it	It's great if it happens, but otherwise not a concern	Reasonably important	Incredibly important	Total
Recycling in hospitals	0.00% 0	0.00% 0	13.64% 6	38.64% 17	47.73% 21	44

This bodes well for Barts Health as there is a clear sustainable motivation that can be harnessed to help improve the environment and waste segregation.

ANALYSIS

From these results it is apparent that the behavioral team's efforts have not been fruitless, however, from my first person fieldwork it is clear that there is room for improvement. The barriers that arose included under-trained staff, as well as a lack of clear labeling and no strategic positioning of bins to ease appropriate sorting of waste. It is clear that while the behavioral change team has improved the overall waste management processes, the 'hug' method is simply not enough for when the staff is unsupervised or under stress.

It is therefore my opinion that a nudge should be implemented to facilitate waste segregation for staff and should address mental shortcuts humans (in this case the bin users) naturally take in their automatic frame of mind.

In order to benefit people, this should facilitate waste management for staff. It is my belief that instead of investing finances into innovations like the solar bins and reverse vending machines, the first step would be to improve waste management where it is most critical. It is my opinion that a nudge with clearer labeling should be piloted at Whipps Cross, where waste management is not as advanced as the other 5 hospitals. Waste management across the

hospitals should ideally be standardized, and by measuring the success of the proposed nudge and scaling it across all six sites, this could be one solution to that barrier. It is imperative that the pilot is tested appropriately in one department in order to ensure that it would be beneficial to implement across Barts Health.

DESIGN

Before prototyping any designs, I outlined the best practices for bins and labeling,³⁶ and in what ways they could nudge better waste segregation.

Firstly, choosing appropriate containers are important. At present, the large bins across Barts Health contain 300L, yet the rest of the bin sizes vary greatly across departments. In addition to this, they should be strategically placed across the sites as to remove any barriers for the users. The domestic and clinical waste bins should be paired together, so there is always the option to use one in equal distance. If they are on their own or set too far apart, the users will not make the extra effort to sort the waste. It is also imperative that in certain departments where specific bins are required i.e. anatomical waste in surgery, that they are available. This increases convenience as well as health and safety.

It is my belief these containers should be streamlined throughout all six sites and to also be color-coded so they become more intuitive to use. Consistency is lacking throughout the sites, and this makes it very difficult to alter behavior. Aside from color-coding the bins, they should also be labeled with clear images and keep text to a minimum. Signs let people know what materials are expected in each collection container, and therefore posters above or on top of the bins would be highly beneficial.

PROTOTYPING

I began the prototype of my designs by observing the current labels at The Royal London, with examples of what waste goes into Clinical Waste and General Waste. From the few labels that were present, the only variation distinguishing the two types of waste were the words "General" and "Clinical". Aside from that, the labels were visually incredibly similar. Both were printed in black and white, had picture images on top and the word "NOTICE". This format does not appeal to the nudge strategy of being attractive, as there is little distinction or contrast that would catch the users eye to be more reflective during their decision making process:

³⁶ "For More Information about Starting a Recycling Program in Your School, Visit Our School Toolkit Page!" *Recycle More Minnesota*. N.p., n.d. Web. 22 Sept. 2015.



DESIGN PROPOSAL

From here I considered the current color labeling of the 8 different types of waste in Barts Health and decided that in order not to add to confusion, I would follow the same scheme. Simplicity in design was key, as I wanted the icons to be intuitive. I chose to focus on streamlining the designs, focus on clarity, contrast and make sure it is cost-effective.

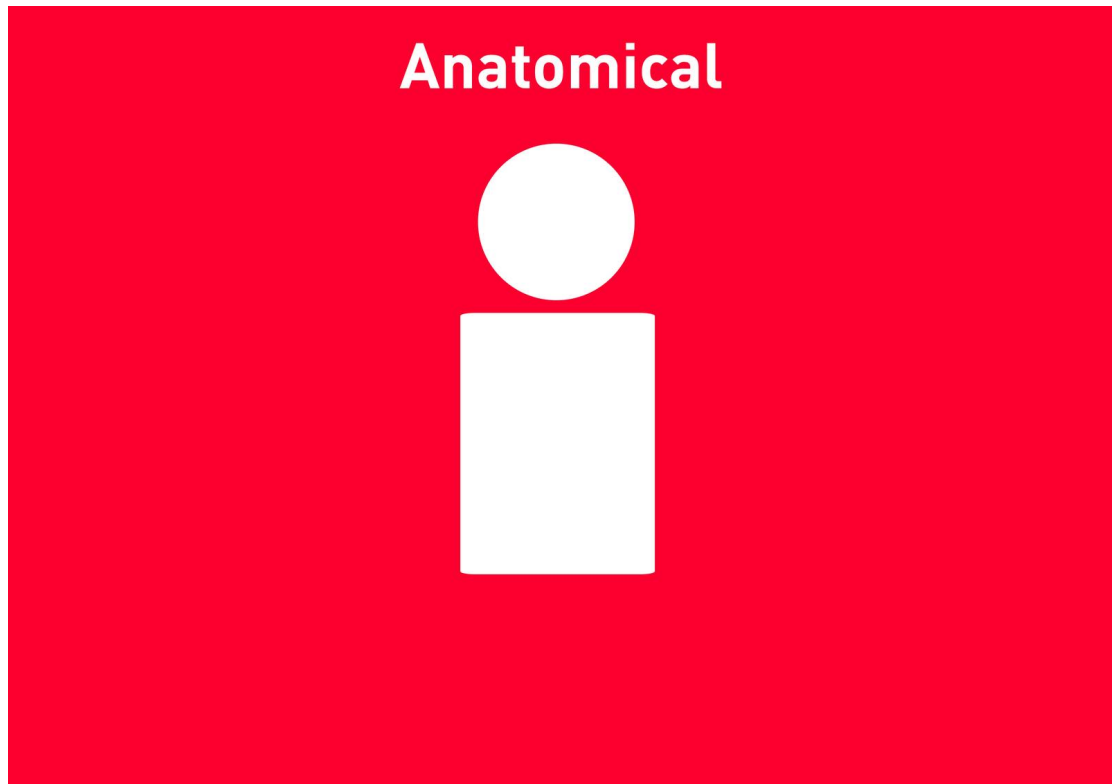
CYTOTOXIC/CYTOSTATIC

This label was the most difficult to illustrate as it encompasses toxic waste from for example chemotherapy. I therefore kept the original cytotoxic label along with its assigned color, purple:



ANATOMICAL

Anatomical waste was also tricky to illustrate as it includes body parts from toes to organs. I felt it was a rather sensitive label to design, so the icon I created is a vague and clinical human form with the assigned red background:



PHARMACEUTICAL

For the pharmaceutical waste, I incorporated the standard pharmaceutical cross on top of the assigned blue background. For the pharmaceutical sharps, however, I altered its assigned color from yellow to blue as to group pharmaceutical waste under the same color-umbrella. For the icon, I designed a plain syringe, which was also used for the blood sharps.

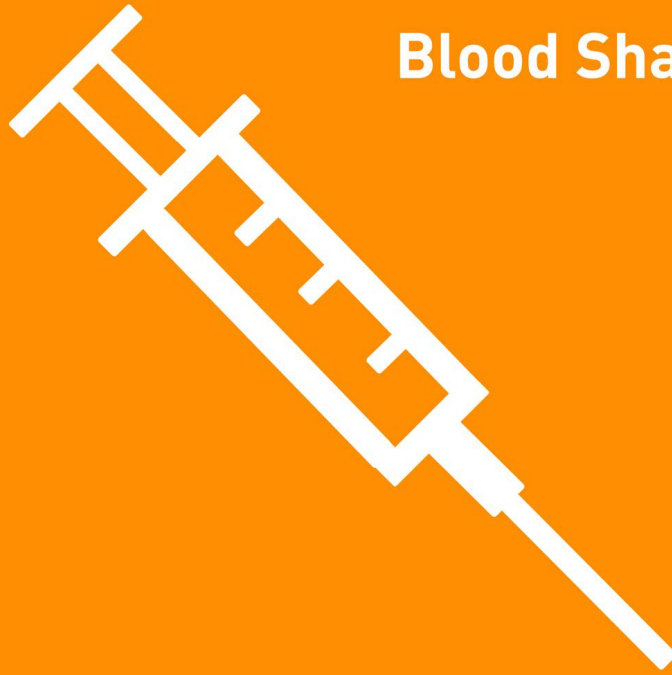
Pharmaceuticals



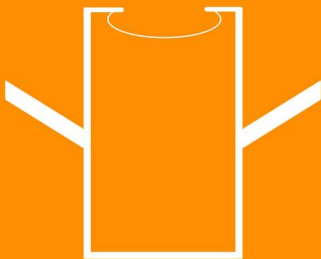
Pharmaceutical Sharps



Blood Sharps



Known Infectious/AT Waste



**£380
tonne**



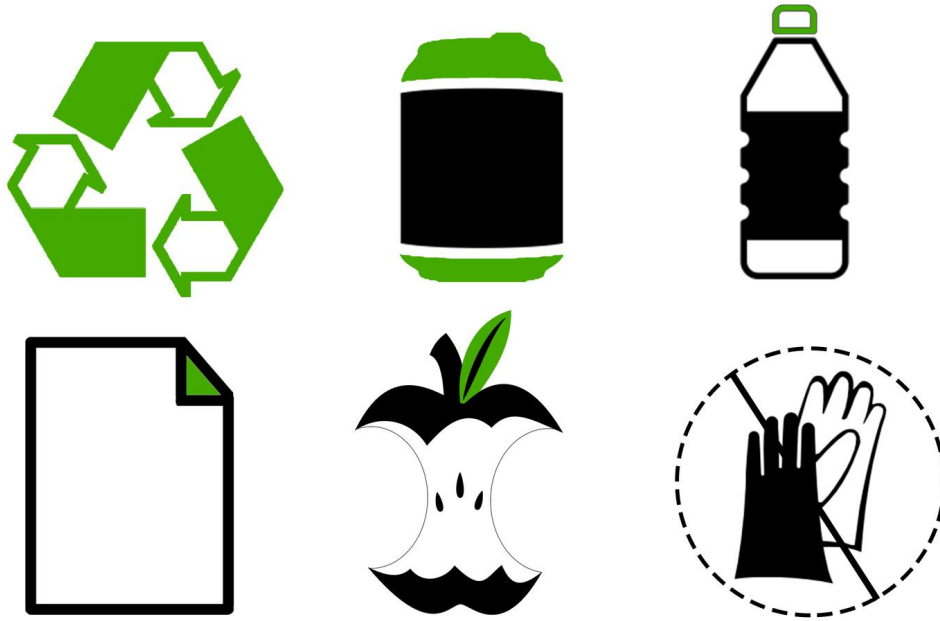
KNOWN INFECTIOUS AND TIGER WASTE

As the same examples of contents go into the known infectious and tiger waste, I kept the same icons of gloves, swabs, aprons, masks and dressings. I did, however, wish to differentiate the two not only by color, but also by the black stripe markings and by indicating the cost of incineration. As the known infectious waste has to be incinerated under higher temperatures as opposed to Tiger waste, which is incinerated, by autoclave, the cost difference is over double. I felt this was important to indicate on the label so that users would be more aware of the financial consequences.

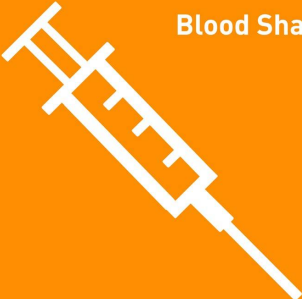
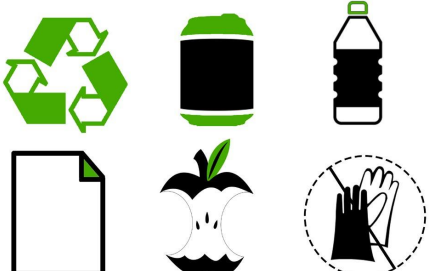
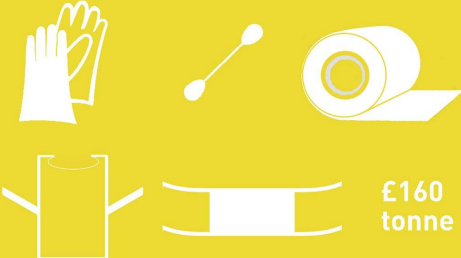
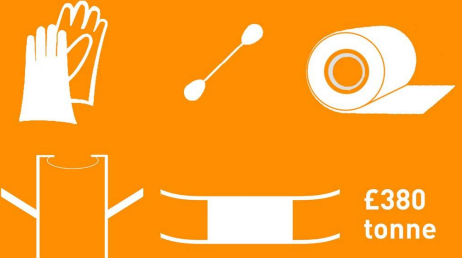
RECYCLING/DOMESTIC WASTE

From my survey, some respondents seemed to be unaware of the fact that all domestic waste is in fact recycled and not disposed of to landfill. I therefore wanted to highlight this fact by adding the recycling label, and green notes on the icons. Aside from the green notes, I kept the icons black on a white background. I chose the white background as the bin bags are clear in the hospitals and the bin tops are black. The icons I illustrated include a can, bottle, paper and an apple core. I added the final icon of the gloves, which should not be placed in the bin, as this is the most common source of cross-contamination in the domestic waste.

Recycling/Domestic Waste



On the following page you can see what the labels look like when they are placed together. It is my hope that the eight variations are easy to distinguish, simple and intuitive to use. These labels could easily be placed on top of the existing bins, meaning that it is simple to put in place and will not be costly.

 Pharmaceutical Sharps	 Blood Sharps
Pharmaceutical Waste 	Anatomical 
Recycling/Domestic Waste 	Cytotoxic/Cytostatic 
 Tiger III Offensive Waste  £160 tonne	Known Infectious/AT Waste  £380 tonne

RESULTS

STAKEHOLDER ANALYSIS

I have conducted a stakeholder analysis in order to evaluate whom Barts Health key stakeholders are, their stakes in the Trust and how they will be impacted from this strategy. This is a useful method as it helps outline what the potential value propositions are for the key stakeholders and therefore the benefits of implementing the strategy.

STAKEHOLDERS	STAKES IN WASTE MANAGEMENT	IMPACT FROM STRATEGY
Barts Health	Financial Environmental-friendliness Reputation Efficiency Legal	- Increased environmental friendliness - Better environment for patients - Less stress for employees - Better reputation - Improved frontline services from financial savings - It will hopefully inspire other Trusts to follow a similar nudge strategy
NHS	Reputation Financial Environment Legal	The media are likely to shed a positive light on environmental practices. There is a chance that there will be a better financial standing for the NHS.
SKANSKA	Reputation Environment Legal Financial	If the strategy is successful, it will reflect well on SKANSKA and their business development will improve. From a financial standpoint, other Trusts may be interested in contracting with them.
Employees	Work environment Wages Stress	- Less stress worrying about a potential sharps injury - Less time-consuming finding correct bins - Gap between their sustainable motivation and actions will be closed - Empower employees - More money used directly back into Trust and less chance of wages being cut
Patients	Hospital environment	- Improved experience through cleanliness and more funding for frontline patient care if strategy is successful financially - Less confusion
Visitors	Hospital environment	- Improved experience - Perception of the Trust will improve - Less confusion
The Government	Financial stakes in NHS Reducing Carbon Footprint	If they could reduce carbon emissions from the NHS, this would benefit the government - It would benefit the government if the NHS saves money - Resources from proper recycling

PESTEL ANALYSIS

In order to gauge the macro-environment and how this strategy could affect hospital waste management in general, I conducted a PESTEL analysis to determine the Political, Economical, Social, Technological, Environmental and Legal implications.

LEGAL:

- Legal implications, protocols to go through
- e.g. lamination, sanitary, correct wordings

POLITICAL:

- Government, funding NHS
- Transparency with public if nudging should be implemented

ECONOMICAL:

- More money to the Trust, which is currently in 93 million pounds deficit
- Cost-effective to implement

ENVIRONMENTAL:

- Significant reduction in CO₂ emissions
- Improved environment for patients
- Re-use of materials that would otherwise be wasted
- Reduced incineration



SOCIAL:

- Mixed feelings with nudging
- Manipulation
- Who nudges the nudgers?
- Crossover with target marketing
- Benefits staff, patients and visitors by decreasing stress and confusion

TECHNOLOGICAL:

- Low-tech solutions as opposed to innovative solar bins or re-vending machines

REFLECTION

In order to assess if the nudge is successful it would be necessary to conduct user testing. Due to the bureaucratic nature of Barts Health and the fact that audits only occur once per month, this testing was beyond the scope of my thesis.

If I were to continue this project, I would propose to implement the nudge in a specific department of Whipps Cross and conduct similar experiments to that of the caramel wrappers used in the Green Footprint Copenhagen example. For the purpose of this experiment, however, I would conduct it over a larger span of time - say for the one month between audits, and focus on solely counting the number of blue medical gloves that are cross contaminated into domestic waste after the labels and strategic placement of bins are put into place.

As I discovered with the TLC program, the next barrier to overcome would be to see how the nudge could scale into the other sites if proven successful. It is my view that implementing a nudge strategy would be more beneficial in the long run as opposed to a shove, smack or the current hug strategy that is in place. The benefit of the nudge compared to the hug strategy is that it requires fewer resources to enforce a behavioral change by appealing to the automatic response. By having the labels permanently in place, this cuts down on the time and effort spent by behavioral team moderating hospital staffs waste segregation. The intervention is also cheaper as no incentives have to be offered. These incentives of cookies and prizes may also wear down in appeal over time.

It is difficult to execute changes in healthcare, as there are so many protocols and laws to consider. In addition, as I discovered from my research of Barts Health, waste containers are not standardized across the six sites, and not even standardized within one hospital. This poses some challenges, but equally highlights the need for change. It is my hypothesis, that if there were significant cost savings benefiting a specific department, this nudge would be attractive for the Trust. It certainly would not cost a significant amount to begin with the rearrangement of existing bins and placement of labels, and to use the funds from cost savings to continue the trial to other departments. In my opinion, it seems counter-productive for a Trust, which is 93 million pounds in debt, to be investing in innovative waste technology like the solar bins, when there are still so many issues prevalent in their current waste management program.

It would be foolish not to harness the staff's motivation to be sustainable through cost-friendly nudges, which facilitate waste segregation. Of course, it must be considered that the nudge might not be as effective as hoped. In that scenario, not too many resources would have been invested in implementing the nudge, nor spent on testing it. It is likely that the cost:benefit ratio will still favor the adoption of the simple solutions proposed by the strategy, and depending on the results, one could potentially refine the nudge to make it more effective. It is my opinion that experimentation through nudges is more worthwhile than doing nothing at all, as the potential pros outweigh the present cons.

CRITIQUE

A critique of Nudging is that some users may find the approach to be manipulative. 'Critics of nudge and libertarian paternalism argue that attempts to change people's lifestyle choice are potentially patronizing and condescending and embody excessive state interference. The use of public authority to change citizens' behaviors, even if the altered behaviors are better for the citizens themselves, violates spheres of privacy, integrity and autonomy... The state ought not to patronize people, by singling out a subset of the population and treating them as if they lacked the full use of reason'.³⁷

In defense of Nudging, Thaler and Sunstein note that 'there is no such thing as neutral design' and as such, the anti-nudge position is unhelpful – a literal nonstarter'. Indeed, 'once you know that every design element has the potential to influence choice, then you either close your eyes and hope for the best, or you take what you know and design programs that are helpful'.³⁸ The important thing to remember about Nudging is that it serves as a signpost to suggest a path towards a particular direction, rather than re-paving the entire road. It's a nudge, not a shove.

Perhaps the true difficulty lies in confusion over what determines a nudge. A nudge, if used correctly, should be; voluntary, avoidable, easy or passive and low cost. In addition, as opposed to other types of behavioral steering such as target marketing, the nudge should always promote the alternative most aligned with the long-term goals of the person who is being nudged. As expressed by Thaler and Sunstein, a nudge should "only get designed to help people live according to their best interests"³⁹. This requires a nudge to be transparent, as the intended outcome as well as its reason should be available to the person being nudged.

That is not to say there are not valid and important ethical concerns raised surrounding the potential misuse of human subconscious. These ethical aspects are important issues for designers to reflect upon when proposing a new contribution to society through design. It becomes especially urgent as the use of design thinking is evolving into realms of solving public and societal concerns.

In a report on health policy, the Behavioral Insights team or "Nudge Unit", which put in place by Cameron in 2010, suggestions of ways to nudge people into making better decisions for their health were proposed. The report argued: "The Government cannot address these issues successfully using heavy-handed legislation to rebalance our diets, change our desire to drink too much alcohol on a Friday night, or make our lives more active. But in a number of areas, there are often cost-effective ways of encouraging behavioral change that

³⁷ Rainford, Paul. *Research Online Website. Innovating through Design in Public Services Seminar Series 2010-2011* (n.d.): n. pag. LSE. Web. 22 Sept. 2015.

³⁸ Thaler, Richard H., and Cass R. Sunstein. *Nudge: Improving Decisions about Health, Wealth, and Happiness*. New Haven, CT: Yale UP, 2008. Print.

³⁹ Thaler, Richard H., and Cass R. Sunstein. *Nudge: Improving Decisions about Health, Wealth, and Happiness*. New Haven, CT: Yale UP, 2008. Print.

are less intrusive and will lead to better results for individuals and for society.”⁴⁰

Using targeted designs to change how people behave has the power to influence social, environmental or cultural issues in positive ways. Dan Formosa of Smart Design believes that designing behavior is the “agenda for the future” as “design is easy, the person is really the problem. We need more psychologists”.

“As choice architects get greater insight into these facts it should be possible to design environments that make choosing for our own greater good a more frictionless, straightforward way of life. When understanding how the human brain functions, it becomes clear that the way our environment is built today, it is often easier to make choices that prevent us from reaching a beneficial behavior, than actually making the right decisions for future wellbeing.”⁴¹

CONCLUSION

It is my hope that this design proposal has the potential to guide users into more sustainable waste management actions. I attempted to provide solutions to the barriers faced by the current waste management strategy in Barts Health by addressing cost, time and the environment by suggesting a nudge, which benefits the triple bottom line.

In essence the proposal is inherently simple and intuitive, however, the brief of a nudge is just that. For such a large institution such as Barts Health, small changes through nudging experiments can have an enormous impact. It is clear due to human nature of automatic and reflective systems, that the hug system to encourage behavioral change is simply not enough.

The suggested nudge of strategically rearranging the bins and labeling them in a clear manner, align with the staff’s motivation to segregate waste and be sustainable. It is through experiments that one would be able to test if it would actually be successful. This nudge would align well with the current waste management strategy, as it would be easy to measure the success of the trials during the monthly audits that are already in place.

In my opinion, it is better to experiment rather than do nothing, especially with matters that concern the environment. Whilst I agree the power of nudging could be used in a negative way, the true definition of a nudge benefits all. When channeled to the positive it seems foolish not to take advantage of human psychology as it can have such tremendous positive effects in such a simple and cost-efficient manner.

Design has the power to influences behavior in ways that are not always well understood by designers. One problem is the lack of awareness of the shared responsibility we all have and the mindset

⁴⁰ Kirkup, James. “Musical Stairs Could Nudge Britain to Better Heath.” *The Telegraph*. Telegraph Media Group, 31 Dec. 2010. Web. 22 Sept. 2015.

⁴¹ Hunnes, Mari. *Nudging How Human Behavior Is Affected by Design*. Thesis. Department of Product Design, Norwegian University of Science and Technology, 2014. N.p.: n.p., n.d. Print.

that our actions do not have the power to affect change. By encouraging positive habits we can ensure sustainability for the future and remind people that every little bit does help.

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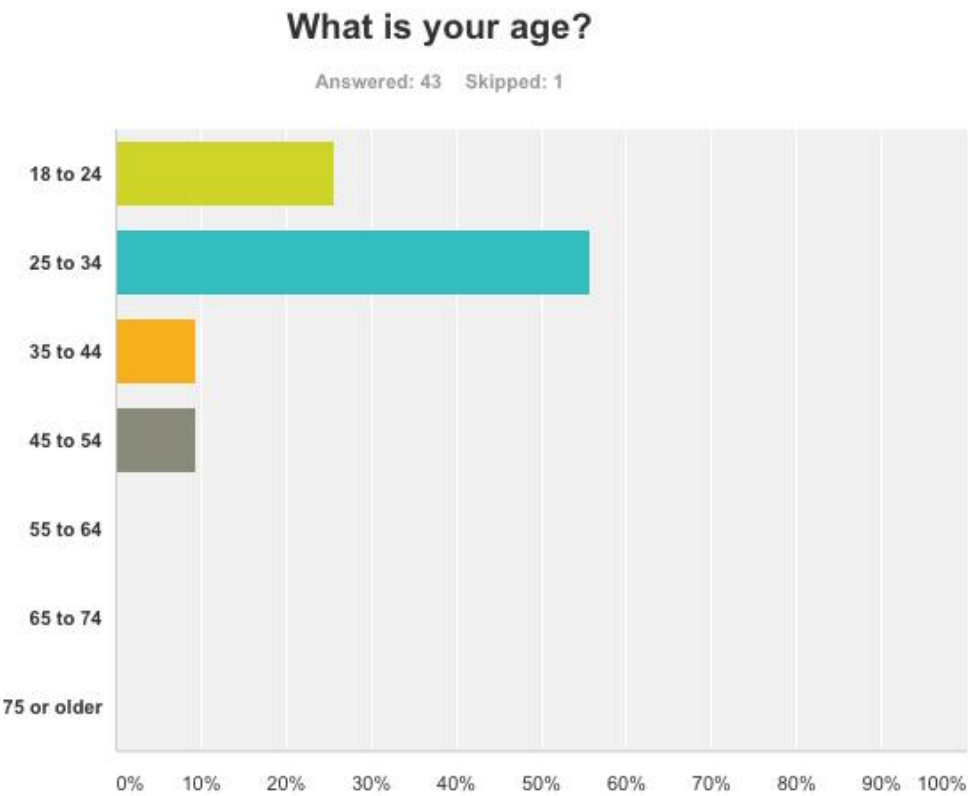
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APPENDIX

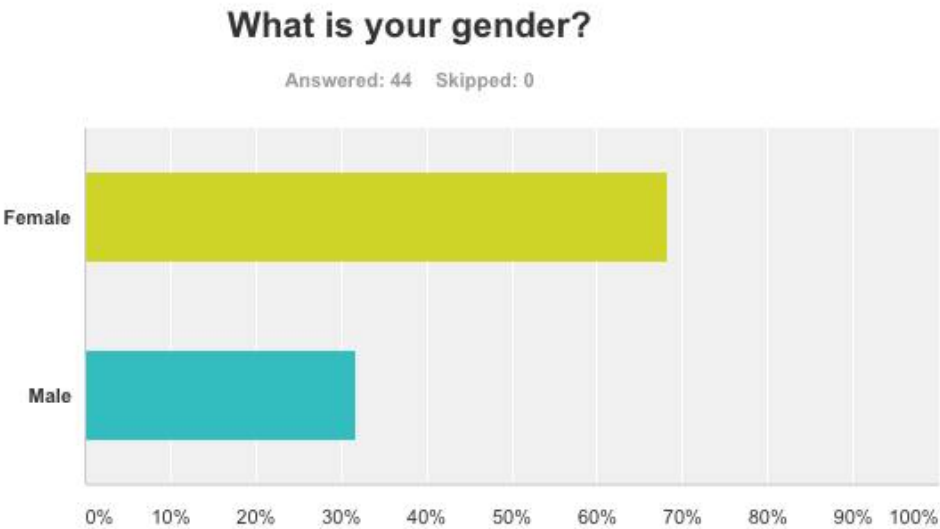
Waste Management in Hospitals

1. What is your age?



2. What is your gender?

- ☐ Female
- ☐ Male



3. At which hospital do you work and in which department?

Whipps Cross
9/14/2015 1:27 PM [View respondent's answers](#)
Whipps Cross
9/14/2015 1:25 PM [View respondent's answers](#)
Whipps Cross
9/14/2015 1:24 PM [View respondent's answers](#)
Whipps Cross
9/14/2015 1:23 PM [View respondent's answers](#)
Various hospitals
9/14/2015 1:21 PM [View respondent's answers](#)
Whipps Cross - Theatres
9/14/2015 1:21 PM [View respondent's answers](#)
Mile End - Community health services
9/14/2015 1:20 PM [View respondent's answers](#)
Whipps Cross
9/14/2015 1:19 PM [View respondent's answers](#)
Whipps Cross - Endoscopy
9/14/2015 1:18 PM [View respondent's answers](#)
Whipps X
9/14/2015 1:16 PM [View respondent's answers](#)
Whipps Crose and Primrose
9/14/2015 1:15 PM [View respondent's answers](#)
Royal London - Pharmacy
9/14/2015 1:14 PM [View respondent's answers](#)
Whipps Cross - Elderly
9/14/2015 1:13 PM [View respondent's answers](#)
Whipps Cross - AAV
9/14/2015 1:12 PM [View respondent's answers](#)
Woodlands and Whipps Cross
9/14/2015 1:11 PM [View respondent's answers](#)
Whipps Cross - Haematology
9/14/2015 1:08 PM [View respondent's answers](#)
Whipps Cross - A&E X-Ray
9/14/2015 1:06 PM [View respondent's answers](#)
Whipps Cross - Radiology
9/14/2015 1:02 PM [View respondent's answers](#)
Royal Cornwall Hospital, Pharmacy
9/10/2015 4:06 PM [View respondent's answers](#)
royal london hospital acute medicine and health care of the elderly
9/10/2015 1:03 PM [View respondent's answers](#)
Southampton General Hospital, Medicine
9/10/2015 12:51 PM [View respondent's answers](#)
Royal London, 9E
9/10/2015 9:06 AM [View respondent's answers](#)
Lillebælt Sygehus (denmark)
9/9/2015 11:49 PM [View respondent's answers](#)
Ain Shams University hospitals, Cairo, Egypt

9/9/2015 9:45 PM [View respondent's answers](#)
ED Addenbrookes

9/9/2015 9:22 PM [View respondent's answers](#)
homerton hospital, elderly care

9/9/2015 7:54 PM [View respondent's answers](#)
William Harvey, Acute Medicine

9/9/2015 7:04 PM [View respondent's answers](#)
Cardiovascular department. IMS Katsushika Heart Centre (in Japan).

9/9/2015 7:00 PM [View respondent's answers](#)
Charing Cross Hospital - medicine

9/9/2015 6:51 PM [View respondent's answers](#)
Homerton, Respiratory Medicine

9/9/2015 5:58 PM [View respondent's answers](#)
Homerton - therapy services

9/9/2015 5:54 PM [View respondent's answers](#)
Whipps Cross General Medicine

9/9/2015 5:41 PM [View respondent's answers](#)
Hospital in Singapore

9/9/2015 5:40 PM [View respondent's answers](#)
Surgery, huddersfield

9/9/2015 5:36 PM [View respondent's answers](#)
Whipps Cross, psychiatry

9/9/2015 5:31 PM [View respondent's answers](#)
Physiotherapy

9/9/2015 5:29 PM [View respondent's answers](#)
St Thomas's acute medicine

9/9/2015 5:16 PM [View respondent's answers](#)
University hospital in Saudi Arabia, infection control department

9/9/2015 5:15 PM [View respondent's answers](#)
Whittington, Medicine

9/9/2015 5:04 PM [View respondent's answers](#)
Southend Hospital, Pharmacy

9/9/2015 4:47 PM [View respondent's answers](#)
Cork university hospital - surgery

9/9/2015 4:36 PM [View respondent's answers](#)
(1) Cardiology (Damansara, Kuala Lumpur) & (1) Administration (Prince Court, Kuala Lumpur)

9/9/2015 4:28 PM [View respondent's answers](#)
I worked at all the barts health hospitals inc the royal London

9/9/2015 7:05 AM [View respondent's answers](#)
Centre Hospitalier de Pau (France), logistics department

9/8/2015 7:46 PM [View respondent's answers](#)

4. Did you have any formal training on waste management before beginning your practice?

- ☐ Yes, it was informative
- ☐ Yes, but only minimal
- ☐ No



5. How important do you consider recycling to be in the hospital?

- ☐ Incredibly important
- ☐ Reasonably important
- ☐ It's great if it happens, but otherwise not a concern
- ☐ Do not consider it
- ☐ Not important

	Not important	Do not consider it	It's great if it happens, but otherwise not a concern	Reasonably important	Incredibly important	Total
Recycling in hospitals	0.00% 0	0.00% 0	13.64% 6	38.64% 17	47.73% 21	44

6. How easy is it for you to find the correct bins?

Not very

9/14/2015 1:27 PM [View respondent's answers](#)

Not very

9/14/2015 1:25 PM [View respondent's answers](#)

Very easy

9/14/2015 1:24 PM [View respondent's answers](#)

Very hard

9/14/2015 1:23 PM [View respondent's answers](#)

Easy

9/14/2015 1:21 PM [View respondent's answers](#)

Very easy

9/14/2015 1:21 PM [View respondent's answers](#)

Very easy

9/14/2015 1:20 PM [View respondent's answers](#)

Near the toilet, sink and corridor

9/14/2015 1:19 PM [View respondent's answers](#)

We always use the signposts

9/14/2015 1:18 PM [View respondent's answers](#)

Navigation can sometimes be difficult

9/14/2015 1:16 PM [View respondent's answers](#)

Easily accessible

9/14/2015 1:15 PM [View respondent's answers](#)

Not very

9/14/2015 1:14 PM [View respondent's answers](#)

Very

9/14/2015 1:13 PM [View respondent's answers](#)

Not too difficult

9/14/2015 1:12 PM [View respondent's answers](#)

Easy to find - different color bins

9/14/2015 1:11 PM [View respondent's answers](#)

It is easy to tell them apart but all options are not always available

9/14/2015 1:08 PM [View respondent's answers](#)

Easy peasy

9/14/2015 1:06 PM [View respondent's answers](#)

Quite easy

9/14/2015 1:02 PM [View respondent's answers](#)

Well labelled bins

9/10/2015 4:06 PM [View respondent's answers](#)

Clinical and general are easy to find. Recycling - imposibl. There r none

9/10/2015 1:03 PM [View respondent's answers](#)

Moderately

9/10/2015 12:51 PM [View respondent's answers](#)

Can be a challenge depending on placement

9/10/2015 9:06 AM [View respondent's answers](#)

Easy to find bins for confidential material and hazardous waste

9/9/2015 11:49 PM [View respondent's answers](#)

If colours are available, it is easy to indicate and use

9/9/2015 9:45 PM [View respondent's answers](#)

Reasonably

9/9/2015 9:22 PM [View respondent's answers](#)

easy

9/9/2015 7:54 PM [View respondent's answers](#)

Difficult. Often nearest bins are significant distance away in clinical area, especially sharps bins. Results in a lot of walking around with sharps.

9/9/2015 7:04 PM [View respondent's answers](#)

I could find a bin per about 5 beds on average.

9/9/2015 7:00 PM [View respondent's answers](#)

Quite.

9/9/2015 6:51 PM [View respondent's answers](#)

very variable

9/9/2015 5:58 PM [View respondent's answers](#)

Variable.

9/9/2015 5:54 PM [View respondent's answers](#)

Not easy

9/9/2015 5:41 PM [View respondent's answers](#)

Easy

9/9/2015 5:40 PM [View respondent's answers](#)

Relatively

9/9/2015 5:36 PM [View respondent's answers](#)

Difficult

9/9/2015 5:31 PM [View respondent's answers](#)

There isn't any recycling bins in the department

9/9/2015 5:29 PM [View respondent's answers](#)

Can sometimes be difficult

9/9/2015 5:16 PM [View respondent's answers](#)

Very easy

9/9/2015 5:15 PM [View respondent's answers](#)

pretty hard actually

9/9/2015 5:04 PM [View respondent's answers](#)

Relatively easy, bins are available on all wards/departments

9/9/2015 4:47 PM [View respondent's answers](#)

Easy

9/9/2015 4:36 PM [View respondent's answers](#)

(1) Easy for Contaminants; not for general recycling (2) No recycling bins; operating theatres were well equipped

9/9/2015 4:28 PM [View respondent's answers](#)

I'm not aware of recycling bins in the hospital

9/9/2015 7:05 AM [View respondent's answers](#)

Very easy

9/8/2015 7:46 PM [View respondent's answers](#)

7. How confident are you that you are disposing of waste correctly?

Varied

9/14/2015 1:27 PM [View respondent's answers](#)

Confident

9/14/2015 1:25 PM [View respondent's answers](#)

Quite confident

9/14/2015 1:24 PM [View respondent's answers](#)

Not very

9/14/2015 1:23 PM [View respondent's answers](#)

Very confident

9/14/2015 1:21 PM [View respondent's answers](#)

Very confident

9/14/2015 1:21 PM [View respondent's answers](#)

Confident

9/14/2015 1:20 PM [View respondent's answers](#)

Very confident

9/14/2015 1:19 PM [View respondent's answers](#)

Very confident

9/14/2015 1:18 PM [View respondent's answers](#)

90% need to check protocols sometimes

9/14/2015 1:16 PM [View respondent's answers](#)

Fairly

9/14/2015 1:15 PM [View respondent's answers](#)

Not sure I am

9/14/2015 1:14 PM [View respondent's answers](#)

Very

9/14/2015 1:13 PM [View respondent's answers](#)

Quite confident

9/14/2015 1:12 PM [View respondent's answers](#)

Fairly confident

9/14/2015 1:11 PM [View respondent's answers](#)

Fairly confident

9/14/2015 1:08 PM [View respondent's answers](#)

Very confident

9/14/2015 1:06 PM [View respondent's answers](#)

Quite

9/14/2015 1:02 PM [View respondent's answers](#)

Very

9/10/2015 4:06 PM [View respondent's answers](#)

I m only confident about disposal of clinical waste

9/10/2015 1:03 PM [View respondent's answers](#)

Quite confident

9/10/2015 12:51 PM [View respondent's answers](#)

Reasonably, think there is probably more I can do

9/10/2015 9:06 AM [View respondent's answers](#)
Very

9/9/2015 11:49 PM [View respondent's answers](#)
not much

9/9/2015 9:45 PM [View respondent's answers](#)
Not really

9/9/2015 9:22 PM [View respondent's answers](#)
fairly

9/9/2015 7:54 PM [View respondent's answers](#)
Confident I am doing as best as can be done with availability

9/9/2015 7:04 PM [View respondent's answers](#)
I was supposed to separate wastes into three types, general wastes, infectious wastes and needles. So, it was not so difficult to dispose wastes appropriately.

9/9/2015 7:00 PM [View respondent's answers](#)
Confident

9/9/2015 6:51 PM [View respondent's answers](#)
moderately

9/9/2015 5:58 PM [View respondent's answers](#)
Fairly confident

9/9/2015 5:54 PM [View respondent's answers](#)
Not very

9/9/2015 5:41 PM [View respondent's answers](#)
Moderately confident

9/9/2015 5:40 PM [View respondent's answers](#)
Relatively

9/9/2015 5:36 PM [View respondent's answers](#)
I know I'm not, but no options

9/9/2015 5:31 PM [View respondent's answers](#)
Not confident

9/9/2015 5:29 PM [View respondent's answers](#)
Very

9/9/2015 5:16 PM [View respondent's answers](#)
Very confident

9/9/2015 5:15 PM [View respondent's answers](#)
not confident because the right bins aren't there!

9/9/2015 5:04 PM [View respondent's answers](#)
Very, I knew where to dispose of certain medicines

9/9/2015 4:47 PM [View respondent's answers](#)
very

9/9/2015 4:36 PM [View respondent's answers](#)
Not very! I disposed most for incineration or autoclave as I was afraid of contaminants and thought better safe than sorry!

9/9/2015 4:28 PM [View respondent's answers](#)
pretty sure I'm not

9/9/2015 7:05 AM [View respondent's answers](#)
Very confident

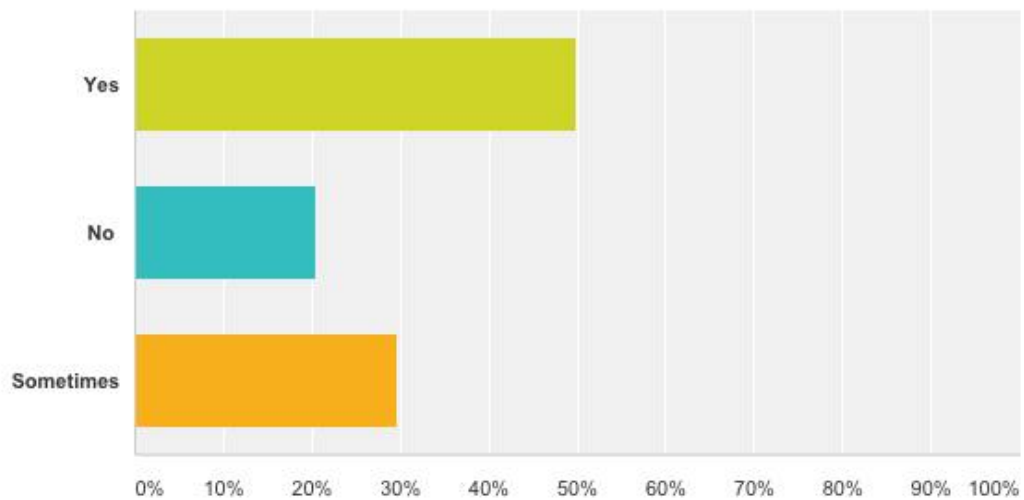
9/8/2015 7:46 PM [View respondent's answers](#)

8. Do you consider the financial impact of cross contamination?

- ☒ Yes
- ☐ No
- ☐ Sometimes

Do you consider the financial impact of cross contamination?

Answered: 44 Skipped: 0



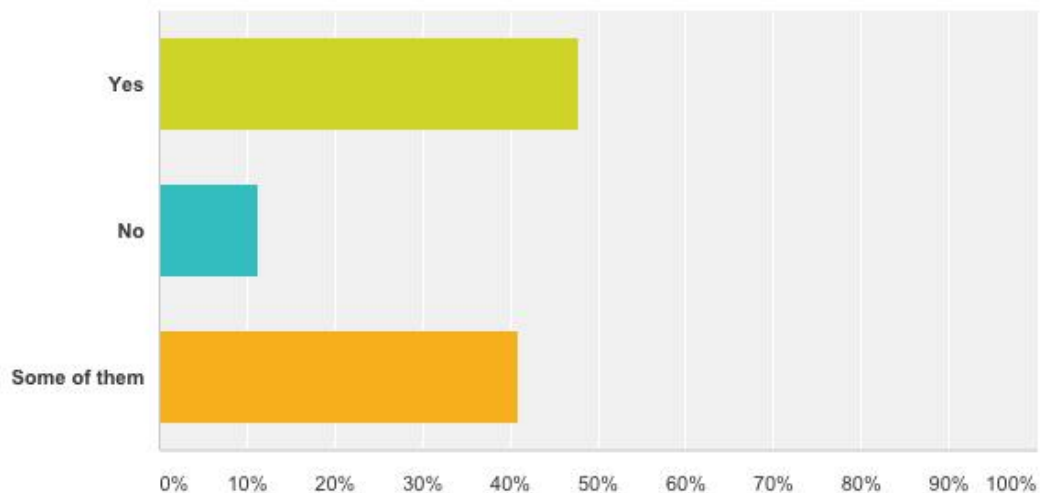
9. Are the labels on the bin clear?

- ☒ Yes
- ☐ No

- ☒ Some of them

Are the labels on the bin clear?

Answered: 44 Skipped: 0



10. What are the greatest difficulties for you in terms of disposing of waste? How do you feel waste management could be improved/facilitated for you?

More bins and different colours

9/14/2015 1:27 PM [View respondent's answers](#)

More waste bins

9/14/2015 1:25 PM [View respondent's answers](#)

More bins and labeling

9/14/2015 1:24 PM [View respondent's answers](#)

Finding a bin, increase number and make them easy to find.

9/14/2015 1:23 PM [View respondent's answers](#)

Do not change the actichlor every day, this is so bad for the environment.

9/14/2015 1:21 PM [View respondent's answers](#)

There are too many bins, reduce them.

9/14/2015 1:21 PM [View respondent's answers](#)

When there is no bin available. Put it in the place where easy to reach to.

9/14/2015 1:19 PM [View respondent's answers](#)

Maybe be put in many places

9/14/2015 1:18 PM [View respondent's answers](#)

Finding the right bin. More visible marking. More education/training

9/14/2015 1:16 PM [View respondent's answers](#)

Icons of what goes into the bin

9/14/2015 1:15 PM [View respondent's answers](#)

Finding domestic waste bins that are clearly labelled for recycling

9/14/2015 1:14 PM [View respondent's answers](#)

Finding the required bin. It could definitely be improved.

9/14/2015 1:12 PM [View respondent's answers](#)

Ensuring patients put waste in correct bins, they just seem to put waste in whatever bin is nearest. Maybe even clearer labeling and colours.

9/14/2015 1:11 PM [View respondent's answers](#)

Availability and increased variety of bins in all areas

9/14/2015 1:08 PM [View respondent's answers](#)

Clearer labeling

9/14/2015 1:06 PM [View respondent's answers](#)

A bit more info from the hospital when joining here

9/14/2015 1:02 PM [View respondent's answers](#)

None, Waste management is taken very seriously in the NHS in particular in a Pharmacy setting due to the hazardous compounds and confidential information

9/10/2015 4:06 PM [View respondent's answers](#)

Addition of recycling bin that r clearly labeled and addition of food waste bin in the canteen would improve the situation. Staff and patient education would improve the situation further

9/10/2015 1:03 PM [View respondent's answers](#)

It is not a major problem for me in my day to day working life

9/10/2015 12:51 PM [View respondent's answers](#)

Maybe more bins, better placement of these, clearer labelling

9/10/2015 9:06 AM [View respondent's answers](#)

confidential bins are sometimes filled up to much

9/9/2015 11:49 PM [View respondent's answers](#)

the availability of indicative bins and labels on them

9/9/2015 9:45 PM [View respondent's answers](#)

Yellow bins everywhere, too few/too varied other bins. I'll dump rubbish into the bin in the room rather than separate it and carry it round the department.

9/9/2015 9:22 PM [View respondent's answers](#)

I suppose a distance between a bin and a patient is one of the largest factors for the difficulties. Ideally, one set of bin should be allocated for one patient.

9/9/2015 7:00 PM [View respondent's answers](#)

Overflowing bins stop people disposing correctly

9/9/2015 6:51 PM [View respondent's answers](#)

colour and positioning of clinical vs offensive.waste - unclear and often appropriate bin not available. however can appreciate some areas will not have space.for all bins!

9/9/2015 5:58 PM [View respondent's answers](#)

More facilities to dispose of/recycle more items.

9/9/2015 5:54 PM [View respondent's answers](#)

Poor labelling and availability of correct bins but also issues with potentially contaminated waste that is also recyclable, not clear what the rules are

9/9/2015 5:41 PM [View respondent's answers](#)

Better labling and incentive

9/9/2015 5:36 PM [View respondent's answers](#)

Appropriate bins

9/9/2015 5:29 PM [View respondent's answers](#)

Closer infectious waste bins, more household waste bins, larger sharps bins, larger IV trays, more signs and pt education

9/9/2015 5:16 PM [View respondent's answers](#)

Education and training

9/9/2015 5:15 PM [View respondent's answers](#)

More bins and recycling!

9/9/2015 5:04 PM [View respondent's answers](#)

I think too much waste is being made in the first place, so for example medicines which have left hospital premises can not be returned and reused, even if they have been returned immediately.

9/9/2015 4:47 PM [View respondent's answers](#)

More bins

9/9/2015 4:36 PM [View respondent's answers](#)

Positioning is not convenient for both medical and admin perspectives; not time-efficient

9/9/2015 4:28 PM [View respondent's answers](#)

To have clearly labelled recycling bins throughout hospital

9/9/2015 7:05 AM [View respondent's answers](#)

The hospital where I work needs to widen its range of types of waste disposal offered (we do not sort enough), and needs to recycle more types of waste.

9/8/2015 7:46 PM [View respondent's answers](#)